Venkata Pushpa Shalini Somboju

Email: venkatapushpashalini@gmail.com
Phone: +1 (470)866-1840

OBJECTIVE

Overall **10+ years** of experience, specializing in cloud platforms including Microsoft Azure and AWS, with strong expertise in DevOps, CI/CD, infrastructure automation, configuration management, cloud migration, and hands-on proficiency with tools such as Docker, Kubernetes, Terraform, Ansible, Jenkins, Git, and Maven.

CERTIFICATIONS

- Microsoft Certified Azure Administrator
- AWS Developer – Associate
- GitHub Actions Certified

ACADEMIC QUALIFICATIONS

- Bachelor of Technology: Sathyabama institute of science and technology - India
  Specialization: Information Technology

PROFILE SUMMARY

- Proficient in Microsoft Azure services such as Azure Kubernetes Service (AKS), App Services, Azure SQL, Azure Active Directory, Azure Functions, Azure Monitor, Azure DevOps, Azure Blob Storage, ADF, ARM templates, and Bicep.
- Experienced in Amazon Web Services (AWS), including EC2, S3, RDS, IAM, Lambda, CloudFormation, EKS, Route 53, VPC, CloudWatch, CloudTrail, and Load Balancers, with a strong emphasis on high availability and scalability.
- Expertise in Infrastructure as Code (IaC) using Terraform, ARM templates, and AWS CloudFormation for provisioning and managing infrastructure across Azure and AWS environments.
- Developed reusable and modular Terraform modules to manage compute, networking (VNETs/VPCs), security groups, Kubernetes clusters, and storage services in a version-controlled, automated fashion.
- Built and maintained CI/CD pipelines using Azure DevOps, Jenkins, GitHub Actions, and GitLab for automated build, test, deployment, and monitoring workflows across multiple environments.
- Configured AWS Lambda functions to execute code in response to HTTP requests through Amazon API Gateway, enabling the creation of serverless APIs.
- Monitored and debugged AWS Lambda functions using AWS CloudWatch Logs and CloudWatch Metrics, ensuring optimal performance and reliability.
- Proficient in containerization and orchestration using Docker, Kubernetes, Helm, and OpenShift; deployed and managed workloads in both Azure AKS and AWS EKS clusters.
- Worked with OpenShift On-Prem Private Cloud Clusters, deploying containers and setting up using the OC CLI.
- Skilled in configuration management and automation using Ansible, Azure Automation DSC, and PowerShell scripting for provisioning, compliance, and state enforcement.
- Implemented end-to-end observability using Azure Monitor, Prometheus, Grafana, ELK Stack, Splunk, and AWS CloudWatch; configured custom dashboards and alerts for proactive incident response.
- Experience with Azure Shared Image Gallery and AWS AMIs for VM image replication and disaster recovery setups; configured Azure Site Recovery (ASR) for BCDR scenarios.
- Strong scripting abilities in PowerShell, Bash, and Python to automate infrastructure, monitoring, and deployment tasks.
- Recognized for collaborating across cross-functional teams to resolve production issues quickly, drive DevSecOps integration, and ensure infrastructure stability, security, and compliance.
- Leveraged Dynatrace's Davis AI to automate incident triage, reduce MTTR, and streamline RCA documentation.
- Enabled SLO-based alerting and dashboards using Dynatrace Grail and Notebooks for business-critical applications.

Area Of Expertise

Technologies

Title	Tools Used
Cloud Environments	Microsoft Azure, Amazon Web Services
Configuration Management	Ansible, Ansible Tower, Chef, Puppet
Build Tools	ANT, Maven, Gradle, sbt
CI/CD Tools	Jenkins, ADO Pipelines, GitHub Actions
Monitoring Tools	Splunk, Nagios, CloudWatch, ELK, Grafana, Prometheus
Container Tools	Kubernetes, Docker
Scripting/Programming Languages	Python, Shell (PowerShell/Bash), Ruby, YAML, JSON, Groovy, JavaScript, C/C++, Shell Scripting,
Version Control Tools	GIT, SVN. Subversion, Bit Bucket, Git Lab
Operating Systems	UNIX, Linux, Windows Server
Databases	SQL Server, MYSQL, Oracle, NoSQL, MongoDB, Dynamo DB, Cassandra
Ticketing Tools	Jira
Testing / Code Quality	Selenium, SonarQube
Web/Application Servers	Apache Tomcat, Nginx, httpd
Virtualization Tools	Oracle Virtual Box, VMWare, vSphere, Vagrant
Infrastructure as Code	Terraform, Cloud formations

Work Experience

Role: Senior DevOps Engineer (Mar 2023– Current)
Client: AT&T

- Worked extensively on Configuring and Provisioning Virtual Machines, Storage accounts, App Services, Virtual Networks, Azure SQL Database, Azure Search, Azure Data Lake, Azure Data Factory, Azure Blob Storage, Azure Service Bus, Function Apps, Application Insights, and Express Route.
- Experienced in utilizing Azure Stack (Compute, Web &Mobile, Blobs, ADF, Resource Groups, Azure Data Lake, Azure Data Factory, Azure SQL, App Services, and Cosmos DB) and services for configuring and deploying Azure Automation Scripts for multiple applications.
- Have worked on setting up Azure Monitor Dashboard for various Azure Services by enabling Diagnostic settings and writing queries in Log Analytics Workspace to send the logs to Azure storage accounts and stream the logs to Azure Event Hubs.
- Hands-on Experience defining Inbound and Outbound rules and associating them with Subnet and Network Interfaces to filter traffic to and from Azure Resources, as well as creating an Azure Key Vault to store certificates and secrets.
- Created and configured HTTP Triggers in the Azure Functions with Application Insights for monitoring and performing load testing on the applications using the VSTS, and used Python API for uploading all the agent logs into Azure blob storage.
- Created and maintained Continuous Integration (CI) using tools Azure DevOps (VSTS) over multiple environments to facilitate an agile development process that is automated and repeatable, enabling teams to safely deploy code in Azure Kubernetes Services (AKS) using VSTS by YAML scripting.
- Worked on integrating Azure-DevOps Boards with Microsoft Teams and Pipelines for Notifying Sprint Boards and Teams, respectively.
- Well-versed in using Terraform templates for provisioning Virtual Networks, Subnets, VM Scale sets, load balancers, and NAT rules. Configured BGP Routes between on-premises data centers and Azure cloud to enable ExpressRoute connections.
- Developed Terraform (TFS) scripts from scratch for developing, Staging, Production, and Disaster Recovery for cloud infrastructure.
- Experience in creating and designing the Terraform (TFS)templates to create custom-sized Resource groups, Kubernetes clusters, containers, blob storage, IOT hub, Event hub. Infrastructure as a code deployment of Web application templates to secure the environment.
- Development of automation of Kubernetes clusters using Terraform (TFS)scripts.

- Performed Azure Scalability configuration that sets up a group of Virtual Machines (VMs) and configures Azure Availability and Azure Scalability to provide High Application Availability and can automatically increase or decrease in response to demand.
- Working on a migration project from On-Premises to Microsoft Azure, contributing to Platform Services Deployment, Including Architecture, Provisioning, Configuration, and Product deployment in the Microsoft Azure cloud platform.
- Developed an automated Stack driver monitoring alerts using Terraform on the Azure Cloud Platform. Worked on Python and PowerShell Runbooks in Automation accounts to build and remove projects within a Subscription.
- Used Azure Kubernetes Service (AKS) to deploy a managed Kubernetes cluster in Azure and created an AKS cluster in the Azure portal using template-driven deployment options such as Azure Resource Manager (ARM) templates and Terraform.
- Used Azure Kubernetes Service (AKS) for implementing ADO pipelines into Azure pipelines to drive all microservices builds out to the Docker images and then deployed to Kubernetes, Created Pods, and managed them.
- Involved in integrating Azure Log Analytics with Azure VMs for Monitoring, storing, tracking Metrics, resolving, and investigating Root cause issues.
- Implemented a Continuous Delivery pipeline with Docker, Microservices, Gitlab, Maven, and Azure.
- Automated CI/CD with ADO Pipelines, build-pipeline-plugin, and Maven. Used JFrog Artifactory.
- Worked with Docker and Kubernetes on multiple cloud providers, from helping developers build and containerize their application (CI/CD) to deploying either on public or private cloud.
- Setting up the complete AKS Kubernetes Dev Environment from scratch to deploy the latest tools which is related to Deep learning and machine learning using helm charts on-premises BareMetal for different teams.
- Deployed multiple microservices into Azure Kubernetes by Dockerizing them and using ADO Pipelines and Azure DevOps.
- Prepared capacity and architecture plan to create the Azure Cloud environment to host migrated IaaS VMs and PaaS role instances for refactored applications and databases.
- Collaborate in the development of main Web Applications to provide invoicing emission services, Responsible of web application deployments over cloud services on Azure, using VS and PowerShell.
- Configuring the Load Balance Sets, Azure Load Balancer, Internal Load Balancer, and Traffic Manager, and working on Application Gateway.
- Worked on PowerShell scripts to automate the Azure cloud system creation of Resource groups, Web Applications, Azure Storage Blobs & Tables, and firewall rules.
- Created and configured Azure Cosmos DB and exposed the service as a Web API.
- Provisioned Azure data lake store and Azure data lake analytics, and leveraged U-SQL to write federated queries across data stored in multiple Azure services.
- Used Ansible and Ansible Tower as a Configuration Management tool to automate repetitive tasks, quickly deploy critical applications, and proactively manage change. Experience in managing Ansible Playbooks with Ansible roles, group variables, and inventory files, copying and removing files on remote systems using the file module.
- PowerShell runbooks were written and deployed to Automation Accounts using CICD Azure DevOps. PowerShell scripts were also written to make API calls to Azure DevOps and find Users who had not accessed Azure DevOps in more than 90 days.
- Implemented centralized container logging and monitoring using Azure App Monitor, Prometheus, and Grafana.
- Configured custom dashboards and alerts in Dynatrace for proactive monitoring of cloud-native microservices, resulting in a 30% reduction in incident response times.
- Leveraged Dynatrace's digital experience monitoring capabilities to track and improve the end-user experience for a Payment application, driving better customer engagement and retention.

Environment: Azure, Terraform, Kubernetes, Azure Boards, Azure repos, Ansible, Shell, Python, Linux, My SQL, Apache Tomcat 7. x, Docker, NoSQL, ARM, Virtualization, Nagios, Splunk, Nginx, LDAP, JDK1.7, XML, Git, Maven, Prometheus, Grafana, Dynatrace.

Role: DevOps Engineer (Jan 2021 – Feb 2023)
Client: American Airlines

- Working on AWS Elastic load balancing for deploying applications in high availability and experience in working on cloud EC2, S3, RDS, Load Balancer, Auto Scaling with AWS command-line interface.
- Using AWS for creating new instances, checking security group settings, adding elastic IPs for servers, deleting the elastic IPs for the servers needed, and applying the inbound IP addresses as needed.
- Created Users, Groups, Roles, Policies, and Identity providers in AWS using Identity Access Management for improvement in login authentication.

- Designed AWS CloudFormation templates to create customized VPC, subnets, and NAT to ensure the successful deployment of Web applications and database templates.
- Created functions and assigned roles in AWS Lambda to run Python scripts, and AWS Lambda using Java to perform event-driven processing. Created Lambda jobs and configured Roles using AWS CLI.
- Configured Ansible control machine & wrote Ansible playbooks with Ansible roles. Created inventory in Ansible for automating the continuous deployment and wrote playbooks using YAML scripting.
- Installed & Implemented an Ansible configuration management system. Used Ansible to manage Web applications, Environment configuration Files, Users, Services, Resources, Mount points & Packages.
- Provisioned the highly available EC2 Instances using Terraform and CloudFormation and wrote new Python scripts to support new functionality in Terraform.
- Deployed Ansible playbooks in an AWS environment using Terraform, as well as created Ansible roles using YAML. Used Ansible to configure Tomcat servers and their maintenance.
- Used Ansible playbooks to set up a Continuous Delivery (CD) pipeline, which primarily consists of Jenkins to run packages and their supporting software components, which are from the Maven build tool.
- Worked on Docker container snapshots, attaching to a running container, removing images, managing Docker file structures, and managing containers. Created packages in containers and committed those containers using both Docker file and Docker commit command, and updated them into the Docker registry.
- Implemented Terraform modules for EC2 Machines, Elastic Load Balancing, EKS Cluster, Azure VM, Azure Data Factory, AKS, and released those modules into GitHub Enterprise Server.
- Integrated Packer with Terraform pipelines for automated image creation and infrastructure rollout.
- Built customized Amazon Machine Images (AMIs) & deployed these customized images based on requirements.
- Using Bash and Python, including Boto3 to supplement automation provided by Ansible and Terraform for tasks such as encrypting EBS volumes backing AMIs.
- Involved in using Terraform to migrate legacy and monolithic systems to Amazon Web Services.
- Provided security and managed user access and quota using AWS Identity and Access Management (IAM), including creating new Policies for user management.
- Created and maintained a cloud application, migrated on-premises application servers to AWS.
- Design AWS CloudFormation templates to create custom-sized VPCs, subnets, and S3 to ensure the successful deployment of Web applications and database templates.
- Configured the Redis Cache service and the File share server in AWS.
- Automate manual infrastructure provisioning and deployment processes to increase efficiency and reduce human error. This may involve scripting and programming using languages like Python, Bash, or PowerShell to automate routine tasks, create custom tools, and streamline infrastructure operations.
- Analyze cloud resource usage, identify cost optimization opportunities, and implement strategies to optimize cloud costs. This may involve rightsizing instances, utilizing reserved instances or savings plans, implementing cost allocation tags, and monitoring cost trends.
- Created Clusters using Kubernetes and worked on creating many pods, replica sets, services, deployments, labels, and health checks using YAML files. Development of automation of Kubernetes clusters via playbooks in Ansible.
- Configured Jenkins, used it as a Continuous Integration tool for installing & configuring Jenkins's master & connecting it with multiple build slaves, and performed Java application builds using Ant & Maven.
- Responsible for installing & administering SonarQube for code quality checks and Nexus repository for generating reports for different projects. Also, integrated SonarQube & Nexus with Jenkins.
- Used build tool Maven for the building of deployable artifacts such as war, jar from source code, and used Nexus for Maven builds. Configured & Administrated Nexus Repository Manager and Artifactory.
- Wrote Python, Ruby, and Bash scripts to monitor the installed enterprise-level applications and manage the configurations of multiple servers using Ansible.

Environment: AWS, AWS Lambda, Kubernetes, Docker, Jenkins 2.0, Apache, Nginx, Ubuntu Linux, VMware ESX, Python, Git, Gitlab, Bash, Ruby, Groovy, Yaml, Packer, SonarQube, Cloud Formation, Terraform, Ansible, Agile/Scrum, SDLC.

Role: DevOps Engineer (Feb 2019 –Dec 2020)
Client: Cox Communications

- Hands-on Experience in creating Azure Key Vaults to hold Certificates and Secrets, designing Inbound and Outbound traffic rules, and linking them with Subnets and Network Interfaces to filter traffic to and from Azure Resources.
- Deployed Azure Cloud services (PAAS role instances) into secure VNETS, subnets, and built Network Security Groups (NSGs) to govern Inbound and Outbound access to Network Interfaces (NICs), VMs, and Subnets.
- Well-versed in automating Infrastructure using Azure CLI, monitoring and troubleshooting Azure resources with Azure App Insights, and accessing subscriptions with PowerShell.

- Configured and maintained Azure Storage Firewalls and Virtual Networks, which use virtual Network Service Endpoints to allow administrators to define network rules that only allow traffic from specific V-Nets and subnets, so creating a secure network border for their data.
- Using Git as an SCM tool with Azure DevOps, I built a local repo, cloned the repo, added, committed, and pushed changes to the local repo, recovered files, set tags, and viewed logs.
- Developed continuous integration and deployment pipelines that automated builds and deployments to many environments using VSTS/TFS in the Azure DevOps Project.
- Focused on using Terraform Templates to automate Azure IAAS VMs and delivering Virtual Machine Scale Sets (VMSS) in a production environment using Terraform Modules.
- Involved in creating Jenkins pipelines to drive all microservices builds out to the Docker images and stores in Docker registry, and then deployed to Kubernetes, Created Pods and managed using Kubernetes, and also performed Jenkins jobs for deploying using Ansible playbooks and Bitbucket.
- Involved in integrating Docker container-based test infrastructure into the Jenkins CI test flow and setting up a build environment integrating with GIT and JIRA to trigger builds using Web Hooks and Slave Machines.
- Written Ansible playbooks, which are the entry point for Ansible provisioning, and in which the automation is described through tasks in YAML format to build a Continuous Delivery (CD) pipeline and execute Ansible Scripts to provide Development servers.
- Data was transferred from On-Premises SQL Database servers to Azure SQL Database servers via Azure Data Factory Pipelines created with the Azure Data Factory copy tool and Self-Hosted Runtimes.
- Generated reports on Jenkins for jobs executed for each channel of business for a period in aiding metrics review.
- Involved in the development of a test environment on Docker containers and configuring the Docker containers using Kubernetes.
- Worked with the OpenShift platform in managing Docker containers and Kubernetes Clusters.
- Created Azure services using ARM templates (JSON) and ensured no changes in the present infrastructure while doing incremental deployment.
- Used Jenkins pipelines to drive all micro services builds out to the Docker registry and then deployed to Kubernetes, Created Pods and managed using Kubernetes.
- Configuring Azure Key Vault services for development teams for handling secrets in dev, test, and production environments using both UI and CLI in Jenkins jobs.
- Provided high availability for IaaS VMs and PaaS role instances for access from other services in the VNet with Azure Internal Load Balancer. Implemented high availability with Azure Classic and Azure Resource Manager deployment models.
- Wrote PowerShell scripts to create the parameter files automatically for all the services in Azure Resource Manager
- Implemented Jenkins pipelines into Azure pipelines to drive all micro services builds out to the Docker registry and then deployed to Kubernetes, Created Pods, and managed using AKS.
- Worked closely with the migration team, which involved developing a technical project plan, reports, recommendations, documentation, landscape design, sizing, testing, and understanding of the hardware platform required for migration.
- Prepared capacity and architecture plan to create the Azure Cloud environment to host migrated IaaS VMs and PaaS role instances for refactored applications and databases.
- Configuring on-prem servers on Jenkins to aid in dev and test deployments for several teams, managing and maintaining credentials on Jenkins.
- Managed Web apps, Configuration Files, mounting points, and packages, launching Azure Virtual Machines (VMs) with Ansible Playbooks.
- Splunk experience includes installing, setting up, and troubleshooting the software, and monitoring server application logs with Splunk to detect production issues.

Environment: Azure, Jenkins, Web logic, Nexus, JIRA, Ansible, Oracle, Terraform, Kubernetes, Prometheus, Python, Maven, Java, GitHub, Linux, ELK, GIT, LDAP, NFS, Splunk, PowerShell Scripts, Shell Scripts, Ansible, Docker, Kubernetes, OpenShift.

Role: DevOps Engineer (Sep 2017 –Jan 2019)
Client: State of Montana

- Configured AWS Route 53 to manage DNS zones globally, create record sets, perform DNS failover and health checks on the Domain, and assign domain names to ELB and CloudFront.
- Configured AWS cloud IAC using Terraform and continuous deployment through Jenkins, and automated the cloud formation using Terraform.

- Created alarms and trigger points in Cloud Watch based on thresholds and monitored the server's performance, CPU Utilization, and disk usage in development and test environments.
- Created Datadog dashboards for various applications and monitored real-time and historical metrics.
- Worked with different instances of AWS EC2, AWS AMIs creation, managing the volumes, and configuring the security groups.
- Created the trigger points and alarms in Cloud Watch based on thresholds and monitored logs via metric filters.
- Used JFrog Artifactory to store and maintain the artifacts in the binary repositories and push new artifacts by configuring the Jenkins project using the Jenkins Artifactory Plugin.
- Designed the data models to be used in data-intensive AWS Lambda applications, which are aimed at doing complex analysis, creating analytical reports for end-to-end traceability, lineage, and definition of Key Business elements from Aurora.
- Worked on the AWS Auto Scaling launch configuration and created the groups with reusable instance templates for Automated Provisioning on demand based on capacity requirements.
- Worked on the AWS IAM service and created the users & groups, defining the policies and roles, and identified providers.
- Worked with the AWS S3 services in creating the buckets and configuring them with logging, tagging, and versioning.
- Designed and implemented AWS EKS Cluster using Terraform from Scratch to implement a fully working EKS Multi-Node Cluster.
- Developed CI/CD system with Jenkins on Kubernetes container environment, utilizing Kubernetes and Docker for the CI/CD system to build, test, deploy, and configure Kubernetes (EKS) to deploy, scale, load balance, and manage Docker containers with multiple name-spaced versions.
- Configured the Jenkins pipeline for creation of Images, pushing them to ECR, and deploying the containerized applications onto Elastic Kubernetes Cluster (EKS).
- Wrote Docker files for microservice applications and good understanding of volumes and Networking for containers.
- Experience in writing YAML files for microservices applications.
- Configured Ansible along with Terraform on Jenkins pipeline to deploy certain application-level dependencies on the Worker Nodes of the EKS Cluster.
- Configured Apache web server and Tomcat in the Linux AWS Cloud environment, and automatic deployment of applications using Ansible automation.
- Involved in writing Java API for Amazon Lambda to manage some of the AWS services.
- Used Jenkins pipelines to drive all micro services, which build out to the Docker registry and then are deployed to Kubernetes by Creating Pods and managed using Elastic Kubernetes Service (EKS).
- Installed and configured ELK stack using HELM charts for monitoring AWS EKS clusters.
- Created custom alerts and notifications in Elasticsearch and Kibana based on CPU or memory usage thresholds, error rates, and latency for AWS EKS clusters.
- Designed and implemented Kubernetes resources, including deployments, services, and persistent volume claims, to deploy and manage applications on OpenShift.
- Creating and managing S3 buckets and implementing access controls for S3 buckets, like bucket policies, ACLs, and IAM roles.
- Involved in designing, deploying, and managing AWS RDS database instances for various applications and environments.
- Responsible for access control for AWS RDS instances using AWS Identity and Access Management (IAM), database users, and database roles to ensure the security of the databases.

Environment: AWS, Jenkins, Ansible, Docker, Docker Swarm, PowerShell, Kubernetes, Terraform, Maven, Ant, SonarQube, Packer, Kubernetes, Python, Datadog, Red Hat, Shell, WebLogic, CI/CD, DynamoDB, Gitlab, Windows, ELK, Git, SVN, Linux, Nexus.

Role: Linux Administrator (Aug 2013 –Sep 2016)
Client: Humana

- Installed and configured RHEL systems using Kickstart, automating deployments with NFS, DNS, DHCP, and NIS setups.
- Performed core system administration tasks including user and group management, memory tuning, and service configuration.
- Managed storage infrastructure using LVM, including volume creation, resizing, and mounting file systems.
- Created and scheduled automated jobs using CRON for routine maintenance and system tasks.
- Configured and maintained Nagios monitoring, writing custom health check scripts for RHEL and ESXi environments.
- Administered virtual machines using VMware ESXi and vCenter, provisioning RHEL, Solaris, CentOS, and Windows servers.
- Troubleshoot system issues by analyzing logs and resolving hardware/software-related errors.
- Participated in disaster recovery planning using LVM-based file systems and Red Hat Clustering.
- Performed kernel and OS upgrades for RHEL and Solaris systems in both production and test environments.